TEXORA · LECTURE SHEET

Weaving Preparatory Process

Yarn as it leaves spinning cannot go straight to the loom. Preparation converts spinning packages into a sized, correctly drawn warp sheet on a weaver's beam — and, for shuttle looms, into weft pirns — so thousands of ends survive shedding, beat-up and abrasion without breaking.

TE-2103 · WEAVING TECHNOLOGY · TOPIC 1 · LEVEL 2 TERM 1

1. Why preparation is needed

A woven fabric interlaces two sets of yarn: the **warp**, running lengthwise, and the **weft**, running crosswise. The warp suffers far more. Every end passes through a drop wire, a heald eye and a reed dent; it is raised and lowered at every pick to form the shed, rubbed by the reed at beat-up, and dragged against neighbouring ends and the weft-insertion element for the whole piece. At several hundred picks a minute, a weak place in an end is tested thousands of times.

Spinning yarn is not ready for that treatment, for four reasons:

- **The package is wrong.** Spinning delivers small bobbins of uneven content. Weaving needs one very large, uniform warp package — the weaver's beam, carrying thousands of ends of equal length and tension — or, for weft, a small package that fits inside a shuttle.
- **The yarn contains faults.** Thick and thin places, neps and weak spots tolerable in spinning cause end breaks, stoppages and cloth defects in weaving.
- **The yarn is too weak and too hairy.** Protruding fibres entangle with neighbours, sheds will not open cleanly, and the end breaks under abrasion. The warp must be strengthened and its hairs laid down — the job of sizing.
- **The ends are not in weaving order.** Each end must occupy an exact position — its own heald, the correct reed dent, the correct place in the colour sequence. That order is built during preparation; it cannot be created on the loom.

Preparation therefore decides, before weaving starts, how fast the loom can run, how often it will stop, and how clean the fabric will be: *weaving is made or ruined in preparation*, because faults built into a beam travel with it for its whole life on the loom.

2. The two streams at a glance

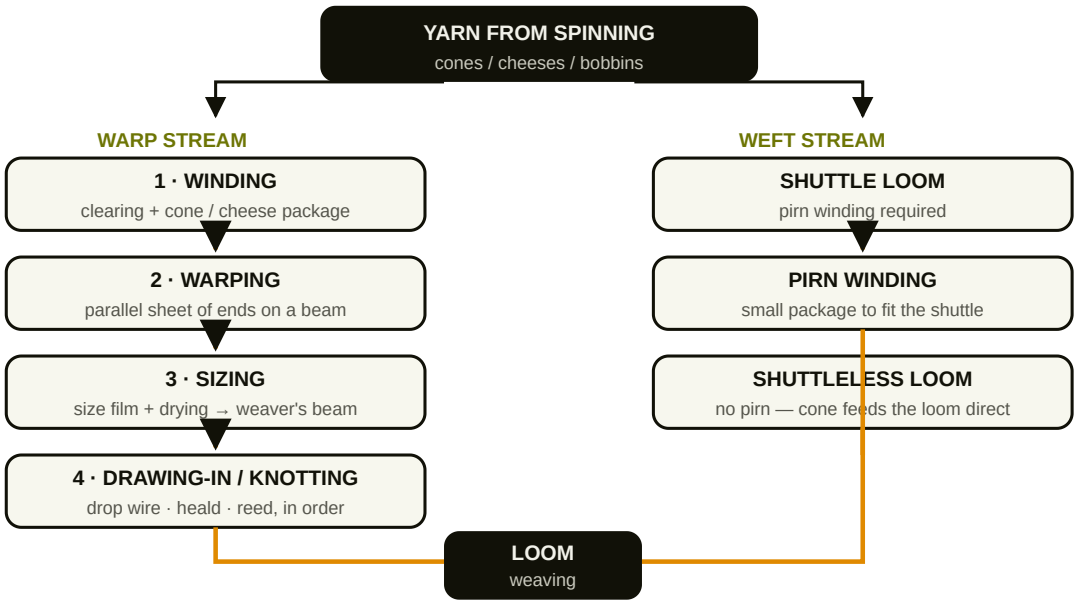


Figure 1 — The weaving preparatory flow. Warp yarn passes through winding, warping, sizing and drawing-in (or knotting) before reaching the loom; weft yarn needs pirn winding only when a shuttle loom is used.

Warp preparation is the main stream: **winding → warping → sizing → drawing-in or knotting → loom**. **Weft preparation** is short: a shuttle loom needs the weft rewound onto a small **pirn**; a shuttleless loom needs none, drawing weft direct from a large cone through an accumulator.

3. Winding — clearing and package building

Winding transfers yarn from the spinning package to a larger, better-built package — a cone or cheese — at high speed, under controlled tension, through a clearer on the way. Its objects:

- **Clearing.** The yarn passes through a **yarn clearer** — mechanical (slot or disc) or electronic (capacitive or optical). Faults outside the set limits — thick and thin places, neps, double ends, foreign matter — are cut out and the ends rejoined by a knot or, better for warp yarn, by a **splicer**, whose joint is almost as slim and strong as the yarn itself; a bulky knot hanging up in a heald eye is a future end break.
- **Package building.** Small bobbins are replaced by one large cross-wound **cone or cheese** for the warping creel, built with a stable traverse so it unwinds smoothly at high speed without sloughing off. Winding also assembles yarn of the required count, colour and lot in one place, and its tension devices even out package density.

Faults missed at winding are never removed later — warping and sizing have no clearing action — and clearer settings balance a cleaner yarn against the joining cycles each cut costs.

> **WORKED EXAMPLE — WINDING PRODUCTION**

A winding drum drives yarn at a surface speed of 1200 m/min. What length is wound per hour, and what weight of 20 Ne (English cotton count) yarn is that?
Length per hour = 1200 m/min × 60 min = 72,000 m/h.
Convert to yards: 72,000 × 1.0936 ≈ **78,740 yd/h**.
For cotton count, one hank = 840 yd, and Ne is the number of hanks in 1 lb, so 1 lb of 20 Ne contains 840 × 20 = 16,800 yd.
Weight per hour = 78,740 ÷ 16,800 = **4.69 lb/h**; in kilograms, 4.69 × 0.4536 ≈ **2.13 kg/h per drum**.
This is the theoretical output at full speed; actual output is lower once clearer cuts, package changes and stoppages are allowed for through machine efficiency.

4. Warping — the sheet of parallel ends

Warping takes the required number of ends from a creel of wound packages and lays them side by side — parallel, evenly spaced, under equal tension — as a flat sheet wound onto a beam. Its objects: to assemble the **exact total number of ends** in the correct colour and count order; to wind all ends at **equal tension and equal length** (a slacker end weaves wavy, a tighter end strained and shiny); and to produce the **required length**, so beam changes on the loom are infrequent. Every warping machine therefore has a creel with a tensioner per package, a guide reed setting sheet width and spacing, an end-break stop motion, and a length-measuring device.

WORKED EXAMPLE — TOTAL WARP ENDS

A fabric is to be woven 150 cm wide in the reed at 28 ends/cm.
Total warp ends = width in reed × ends per cm = 150 × 28 = **4200 ends**.
Check: 100 × 28 = 2800 and 50 × 28 = 1400; 2800 + 1400 = 4200 ends. (Selvedge ends are added to this ground total in practice, and the result is matched to the creel and warping plan.)

4.1 The two warping systems

FEATURE	BEAM (HIGH-SPEED / DIRECT) WARPING	SECTIONAL WARPING
Principle	Several hundred ends are wound as a partial-width sheet onto a warper's beam; several such beams are combined at sizing to give the full end count.	The full end count is built section by section on a tapered drum, each section at full fabric density and width; the completed sheet is then transferred to the weaver's beam.
Typical creel	About 400–800 ends per beam	About 300–600 ends per section, many sections

Best for	Long runs of plain, single-colour warps	Short runs, striped and patterned warps, fine and dyed yarns
Pattern accuracy	Colour order set on the creel; limited for complex patterns	Excellent — each section is built to the exact repeat

5. Sizing — the protective film

Sizing (slashing) applies a protective coating, the **size**, to the warp sheet. The sheet passes through a size box of hot, cooked size liquor (traditionally starch-based; modern recipes add synthetics such as PVA and acrylics, plus lubricants), is squeezed to control how much liquor it carries, dried over heated cylinders or by hot air, and wound — all warper's beams combined — onto the **weaver's beam**, the ends kept separated so none sticks to its neighbour. Its objects:

- **A protective film.** Dried size coats each yarn and binds the surface fibres down: hairiness falls and sheds open cleanly.
- **Strength.** Size penetrates between fibres and binds them together; tensile strength commonly rises by roughly 10–30 % for spun cotton warp.
- **Abrasion resistance** — sizing's chief purpose. The film takes the rubbing of heald, reed and shuttle in place of the fibres. The film must be strong yet flexible and slightly elastic: a hard, brittle film cracks and powders off (shedding, or dusting) and protects nothing.
- **Preserved elasticity.** The warp must still stretch and recover at each shed change; over-sizing or harsh drying makes it stiff and *increases* breaks.

The amount applied is expressed as **size add-on**:

$$\text{Size add-on (\%)} = (\text{dry size on yarn} \div \text{unsized yarn weight}) \times 100$$

Add-on is controlled by liquor concentration, squeeze pressure and sheet speed, and checked by desizing a weighed sample. Typical values run from about 5 % for strong or filament warps to 15 % or more for weak, fine spun warps, set to the minimum that keeps breaks low: excess size stiffens the warp and makes desizing harder. Strong filament warps may be woven unsized.

6. Drawing-in and knotting

Before weaving, each end on the weaver's beam must be threaded through three elements:

1. **Drop wires** — one wire rides on each end; if the end breaks or slackens, the wire drops and triggers the warp stop motion.

2. **Healds** — each end passes through the eye of one heald on one shaft. Which shaft an end belongs to is fixed by the weave plan, the **draft**, so drawing-in is where the pattern is physically built into the loom. Correct **heald selection** is therefore a quality decision, not mere threading.
3. **The reed** — ends are drawn through the reed dents at a set number per dent (**denting**), which fixes the ends per centimetre in the cloth and beats up each pick. Correct **reed selection** and denting set the fabric's warp density and cover.

Drawing-in — by hand or automatic machine — is needed whenever the draft or denting changes; one wrongly drawn end repeats as a fault for the whole beam. When the new beam repeats the old draft and denting *exactly*, it is avoided by **knotting** (tying-in): a knotting machine ties each new end to the corresponding old end, and the knots are drawn through the existing healds and reed. Knotting a different-pattern warp is a costly error — the new beam then weaves the old pattern.

7. Weft preparation — pirn winding, and when there is none

On a **shuttle loom** the weft travels inside a shuttle, so it must be carried on a small **pirn** fitting the shuttle chamber. Yarn is rewound on an automatic **pirn winder**, the pirn built with a shaped chase and bunch so it holds the maximum length, unwinds without snatching, and can be changed by an automatic loom without stopping. Weft is not sized, and its faults appear directly across the cloth as bars or thin places.

On **shuttleless looms** there is no pirn: weft is drawn from a large stationary cone through an accumulator storing a few picks under even tension, then inserted by projectile, rapier, air or water jet. Weft preparation disappears entirely.

ITEM	WARP PREPARATION	WEFT PREPARATION
Stages	Winding, warping, sizing, drawing-in / knotting	Pirn winding (shuttle looms only); none for shuttleless
Sized?	Yes, normally	No — inserted once, under low tension
Final package	Weaver's beam — thousands of ends in one package	Pirn, or a cone at the loom side
Main risk if badly done	End breaks, stoppages, lengthwise faults	Weft breaks, missing picks, crosswise bars

8. Quality consequences of bad preparation

- **End breaks.** Uncleared weak places, bulky knots, under-sized or over-dried warp and ends stuck together after sizing all break under shedding and beat-up. End-break rate is the most watched figure

in a weave shed.

- **Loom efficiency falls.** Every stop for a warp or weft break consumes loom time, and a weaver can tend fewer looms when breaks are frequent, raising labour cost per metre as well as losing output.
- **Fabric defects repeat.** A wrongly drawn, missing, slack, tight or out-of-order end runs the full length of the piece — lengthwise faults are the signature of bad preparation and can downgrade a whole beam of cloth.
- **Size faults show at once.** An under-dried, sticky beam causes clusters of breaks at its start; brittle over-sized warp dusts at the reed and breaks increasingly as the beam weaves down.

So quality is measured at preparation — splice quality, warping tension, size add-on, draft — because once a fault shows on the loom, a whole beam carries it.

Glossary

Winding — rewinding yarn onto a larger package while clearing faults. **Clearer** — device detecting and cutting out faults beyond set limits. **Splice** — a slim knotless joint of intermingled fibres. **Warping** — assembling the required ends in parallel, at equal tension, onto a beam. **Sectional warping** — building the warp in bands on a drum, for patterned and short warps. **Sizing** — coating the warp sheet with a protective size film onto the weaver's beam. **Size add-on** — dry size as a percentage of unsized yarn weight. **Drawing-in** — threading each end through drop wire, heald and reed in draft order. **Knotting** — tying a new warp end-for-end to an old warp of identical draft. **Pirn** — the small weft package inside a shuttle. **End break** — breakage of one warp end during weaving.

Key points

• Preparation exists because spinning yarn is on the wrong package, contains faults, is too hairy and weak for weaving abrasion, and is not yet in weaving order.
• Warp stream: winding → warping → sizing → drawing-in or knotting → loom. Weft stream: pirn winding for shuttle looms only; shuttleless looms feed weft direct from cones.
• Winding clears faults and builds the package; warping builds a parallel sheet of the exact end count at equal tension and length — beam warping for long plain runs, sectional warping for patterns and short runs.
• Sizing lays a flexible protective film on the warp: strength and abrasion resistance rise, hairiness falls; the amount is measured as size add-on (%) and set to the minimum that controls breaks.
• Drawing-in builds the weave pattern into the loom through heald and reed selection; knotting replaces it only when the new warp repeats the old draft exactly.
• Bad preparation means end breaks, lost loom efficiency and lengthwise faults repeating for the whole beam.

Practice

1. A fabric is woven 120 cm wide in the reed at 32 ends/cm. Calculate the total number of ground warp ends required.
2. A winding head runs at 1000 m/min on 30 Ne cotton yarn. Calculate the theoretical winding production in lb/h, showing the conversion from metres to yards.
3. A sized warp sample weighs 112 g; after desizing, the same length weighs 100 g dry. Calculate the size add-on and judge whether this warp is lightly or heavily sized.
4. A mill changes from a plain grey warp to a short run of multi-colour striped warp. Which warping system should be chosen, and why?
5. End breaks rise sharply in the first hour after a new beam is gaited, then settle. Give three preparation-side causes fitting this pattern, with a check for each.
6. Explain why weft yarn is never sized although warp yarn of the same count must be, referring to what each yarn does on the loom.

Answer key

1. Total ends = 120 × 32 = **3840 ends**. Check: 100 × 32 = 3200; 20 × 32 = 640; 3200 + 640 = 3840.
2. Length per hour = 1000 × 60 = 60,000 m/h; in yards, 60,000 × 1.0936 = 65,616 yd/h. Length per lb of 30 Ne = 840 × 30 = 25,200 yd. Production = 65,616 ÷ 25,200 = **2.60 lb/h** (≈ 1.18 kg/h) per head, theoretical.
3. Dry size = 112 − 100 = 12 g. Add-on = (12 ÷ 100) × 100 = **12 %** — medium-to-heavy sizing, as a spun warp needing strong protection would carry; a filament warp would need far less.
4. **Sectional warping**: the full colour order is built section by section at final density, so a complex stripe repeat is set accurately, and a short run does not justify a large beam-warping creel with several beams to combine at sizing.
5. Any three: (i) sticky beam from under-dried size — check the drying record and whether ends separate cleanly; (ii) ends crossed or stuck at the lease — inspect the first metres of sheet behind the reed; (iii) mixed-tension ends from warping — check tension records and look for slack or tight ends; (iv) knotting-in knots catching in healds — check knot size. The pattern points to a fault at the start of the beam, not a uniform sizing fault.
6. The warp is tensioned, shed and abraded by drop wire, heald, reed and insertion element at every pick — thousands of cycles per metre — so it needs a size film's strength and abrasion resistance. The weft is inserted once, under low tension, and never worked again; sizing it would add cost and stiffness against a load it never bears.